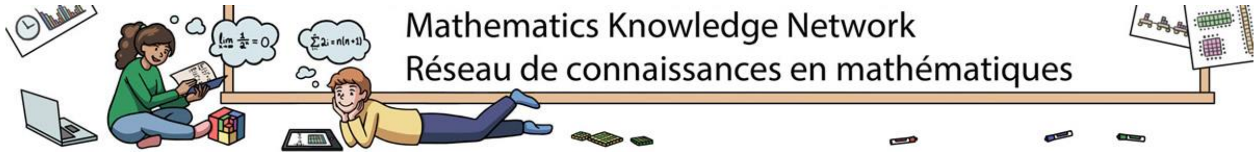

WEEK 16 – Interpreting Graphs

Grade: Intermediate (9) **Unit:** Linear relations

Curriculum Expectations

MPM 1D/MFM 1P: describe the effects on a linear graph and make the corresponding changes to the linear equation when the conditions of the situation they represent are varied

SEL: make connections between math and everyday contexts to help them make informed judgements and decisions

Activity

1) For this activity, you will be given a series of graphs, and your task is to match the graphs with what they represent in their real-world representations.

2) First, you will analyze the graph presented to you. Using the formula below, you will calculate the slope of your line

3) You will, pick two points that your line touches and label them (x_1, y_1) for your first point and (x_2, y_2) for your second. Then, using the equation $(x_2 – x_1)/(y_2 – y_1)$ determine the slope of your line

4) You then choose the appropriate definition from the definition list that correlates to the graph that you are analyzing. Pay particular attention to the slope, y-intercept etc.

5) Lastly you will make one change to the graph. This change can be a change in the y-intercept, the direction of the graph, or a change to the slope of the graph. You will change the definition that you chose to correspond to your changed graph.

Check for Understanding

I understand how linear relations represent real life relationships

I can calculate the slope of a linear relationship by using the rise/run relationship

I understand the relevance of graphs in my everyday life

Materials

Recording sheet (attached below), pencil, information sheet below or internet access, calculator

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Definition List

The rate of interest on a Guaranteed Investment Certificate (GIC) per 4 months

The cost of a cell phone plan per month on a bring your own phone plan

The rate of the cost of a cell phone plan per month with a free phone

The amount of money in a chequing account after hours in a mall

The height of a thrown ball in feet from the ground per second

Heat of water in ^{0}C in a pot on the stove in minutes

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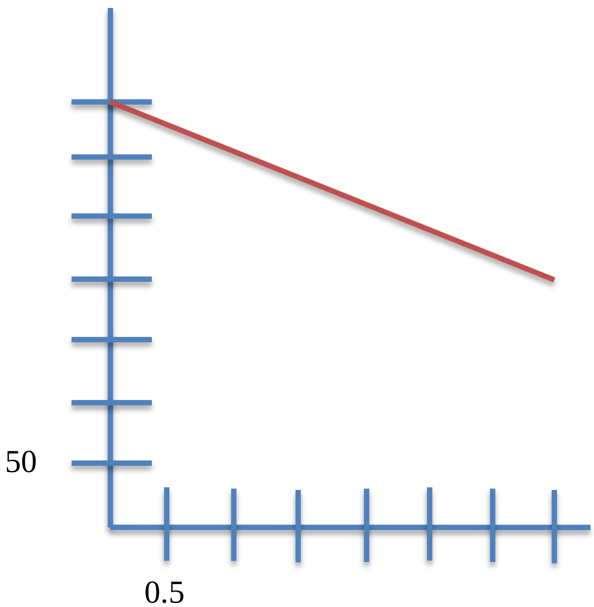
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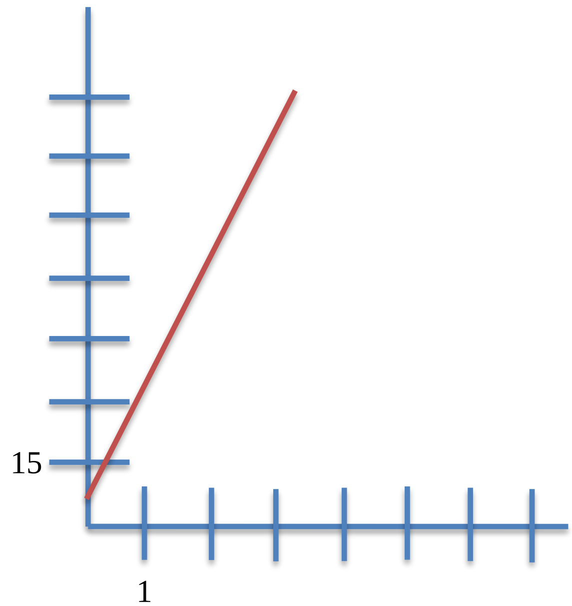
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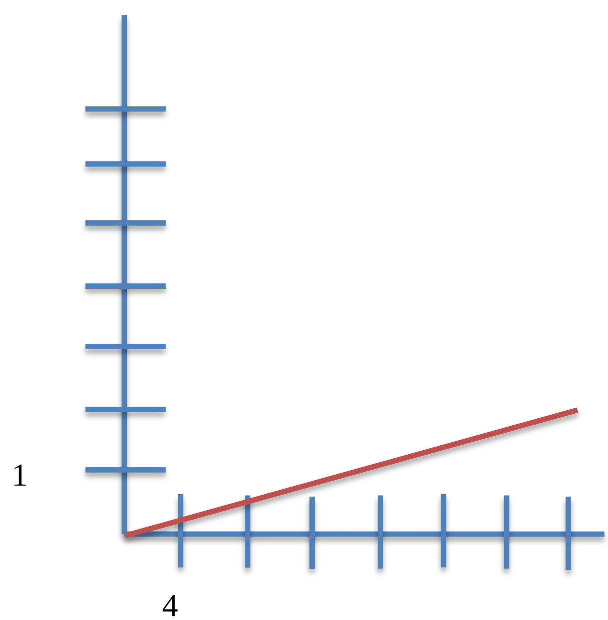
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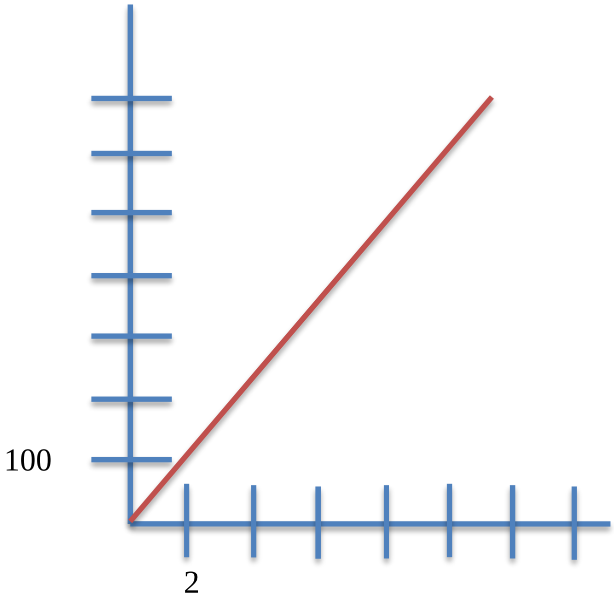

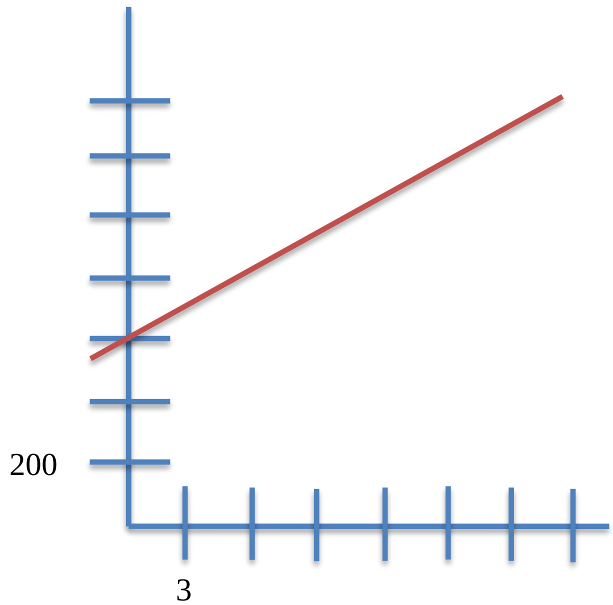

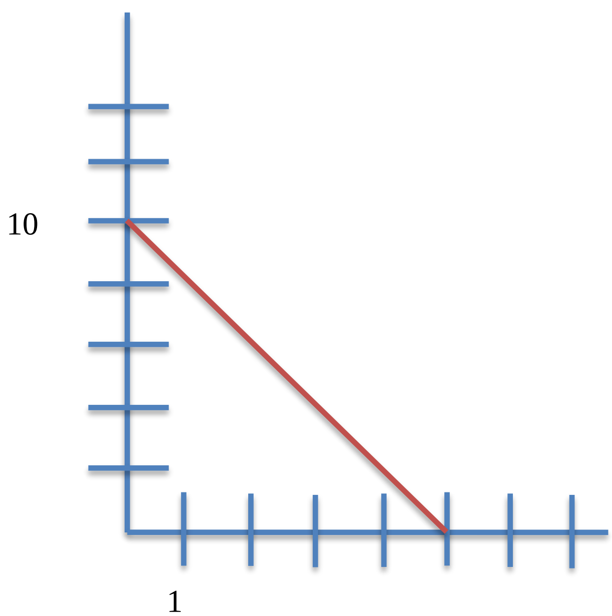

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